

# **DETECTION OF BRAIN TUMOURS USING CONVOLUTIONAL NEURAL NETWORKS**

**A PROJECT REPORT**

*Submitted by,*

**Ms. K Seetha Thanooja - 20211CBD0054**

*Under the guidance of,*

**Mr. Ramamurthy ketha**

*in partial fulfillment for the award of the degree of*

**BACHELOR OF TECHNOLOGY**

**IN**

**COMPUTER SCIENCE AND TECHNOLOGY (BIG DATA)**

**At**



**PRESIDENCY UNIVERSITY**

**BENGALURU**

**JANUARY 2025**

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## **SCHOOL OF COMPUTER SCIENCE ENGINEERING**

### **CERTIFICATE**

This is to certify that the Project report “**DETECTION OF BRAIN TUMOURS USING CONVOLUTIONAL NEURAL NETWORKS**” being submitted by “Seetha Thanooja Kalla” bearing roll number 20211CBD0054 in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Technology (Big Data) is a bonafide work carried out under my supervision.

**Mr. Ramamurthy Ketha**

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### **DECLARATION**

We hereby declare that the work, which is being presented in the project report entitled “**DETECTION OF BRAIN TUMOURS USING CONVOLUTIONAL NEURAL NETWORKS**” in partial fulfillment for the award of Degree of **Bachelor of Technology** in Computer Science and Technology (Big Data), is a record of our own investigations carried under the guidance of **Mr. Ramamurthy Ketha**, Assistant Professor, School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

**NAME**

**ROLL NUMBER**

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# ABSTRACT

The growing incidence of brain tumors requires proper diagnostic tools for early detection and treatment. This project proposes a strong framework for the effective and high quality detection of brain tumors using Convolutional Neural Networks (CNNs), implemented through Google Colab and PyCharm. The proposed model uses advanced deep learning techniques to analyze Magnetic Resonance Imaging (MRI) scans, aiming to improve accuracy and efficiency in tumor identification. It starts by the collection of the comprehensive dataset MRI images that represent different types of brain tumors including meningioma, glioma, and pituitary. Preprocessing is of utmost importance where normalization, resizing, and data augmentation of images are carried out to make it more robust in the model.

The CNN model is trained on a labeled dataset, where features are extracted through convolutional layers that capture spatial hierarchies in the images. The training is done in Google Colab using its powerful GPU capabilities to process large datasets. PyCharm is used as the IDE for code development and testing, providing an opportunity to make iterative improvements and debug the code. Accuracy, sensitivity, specificity, and Area Under the Curve (AUC) are some of the metrics used to evaluate the performance of the model. The CNN implementation for the detection of brain tumors represents a great leap in medical imaging analysis. Using Google Colab for training and PyCharm for development streamlines the workflow so that researchers can focus on optimizing model parameters and enhancing performance. This automated detection system not only helps in timely diagnosis but also reduces the dependency on specialized radiological expertise, making it a valuable asset in clinical settings.

## ACKNOWLEDGEMENT

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